

Cybersecurity

The Cybersecurity Mindset

Lesson 1.1

All graduates in good standing will receive a CompTIA Exam Voucher of Choice and accompanying CompTIA CertMaster Practice Tool.

The **CompTIA CertMaster Practice Tools** is an adaptive knowledge assessment and certification training companion tool that helps you prepare for the CompTIA exam.

- It features question-first design, real-time learning analytics, and content refreshers to reinforce and test what you know and close knowledge gaps.
- You will receive access upon graduation.

All graduates in good standing will receive a CompTIA Exam Voucher of Choice and accompanying CompTIA CertMaster Practice Tool.

CompTIA Exam Voucher for one of the following exams:

- Security+, Network+, Linux+, Server+, or Cloud+
- Your Career Director will be available to help you decide which voucher is best for you based on your skill set and professional goals.
- Upon graduation, you will receive an email to request your exam voucher.
- Vouchers will be sent via email and are valid for 12 months.



Learning Objectives

By the end of today's class, you will be able to:

- 1 Explain the course structure and general direction of the program.
- 2 Recognize the high-level security strategies and tools covered in class.
- 3 Define cybersecurity as the assessment of threats and the mitigation of risk.
- 4 Give a clear definition of the CIA triad and its elements.



The Rising Cyber **Threat**

Why is **cybersecurity** such a
desired skill today?

Why is **cybersecurity** such a desired skill today?

01

Explosive growth in dependence on IT systems

02

More users (targets) on connected devices

03

Better tools for bigger damage

04

Significant investment and higher organization by bad actors

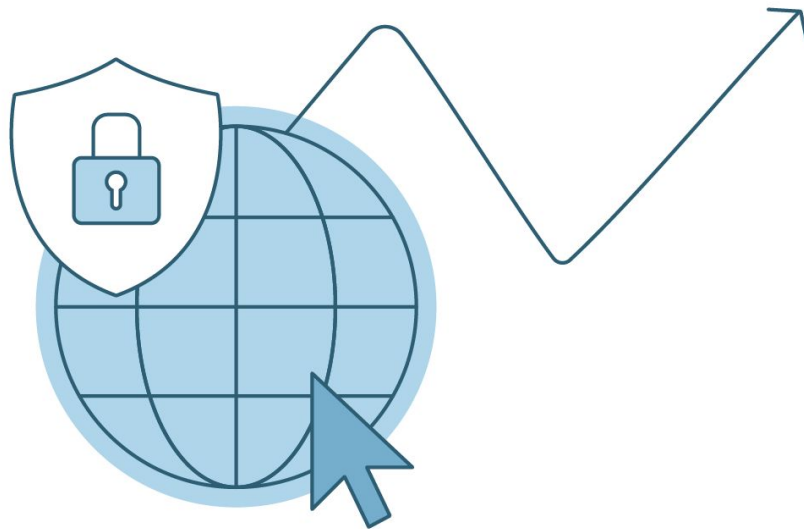
05

Global cybersecurity skills shortage



Explosive Growth

- Personal/social
- Artificial intelligence
- Data protection
- Teleworking
- Security automation





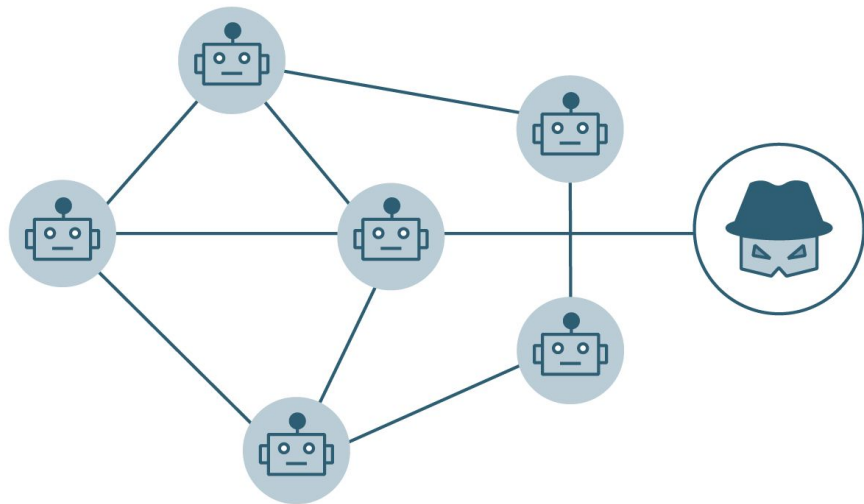
Increased User Base

- Exponential growth in users
- Rise of mobile devices
- IoT/smart homes
- Wearable technology





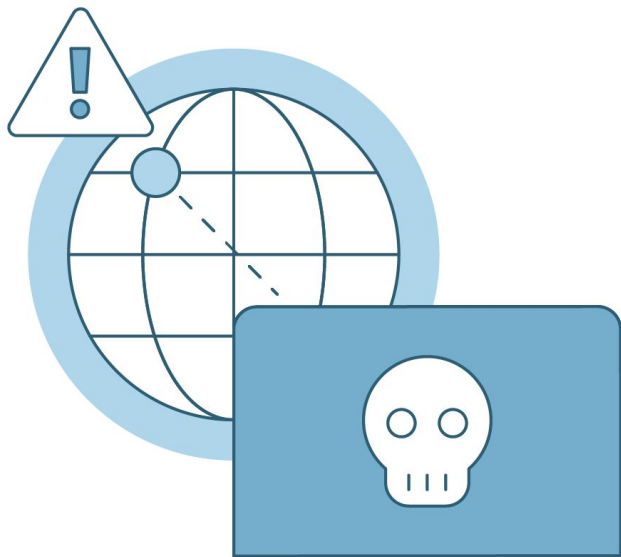
Better Tools



- Automated attack tools
- Sophisticated malware
- Encryption methods



Organized Cybercriminals



The field was once populated by individual **“lone attackers.”**

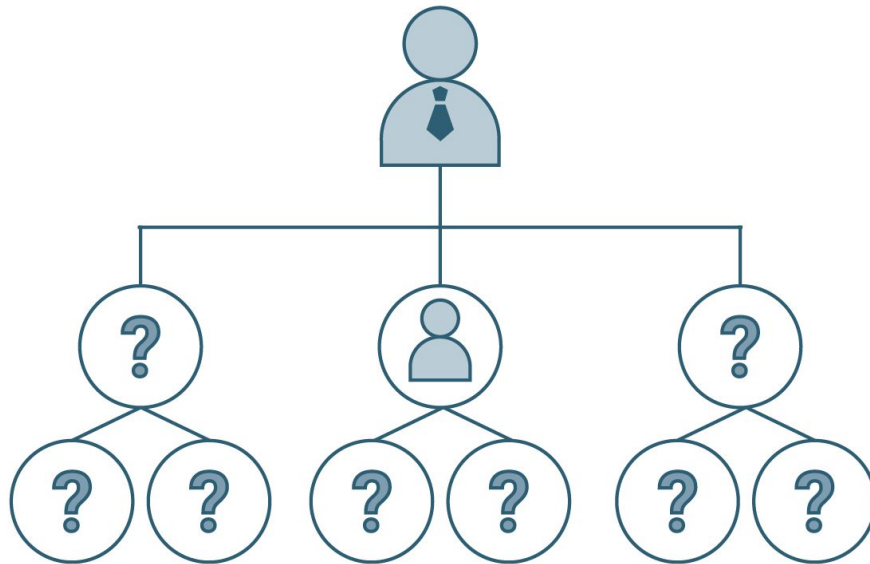
It has now become a focal point for :

- Organized crime
- Nation states
- Private enterprises



Global Shortage

- High demand, low supply
- Rapid technological changes
- Growing cyber threats
- Emergence of new roles





Global Shortage



An **estimated 4 million** professionals are needed to fill the growing cybersecurity workforce gap. At the same time, Fortinet's 2024 Global Cybersecurity Skills Gap Report found that **70%** of organizations indicated that the cybersecurity skills shortage creates additional risks for their organizations.

-Fortinet
June 2024

Emerging Threat Market

Threat landscape is constantly changing and evolving with newer threats, including:



AI-powered attacks



Deepfake technology



Supply chain attacks



Ransomware as a service (Raas)



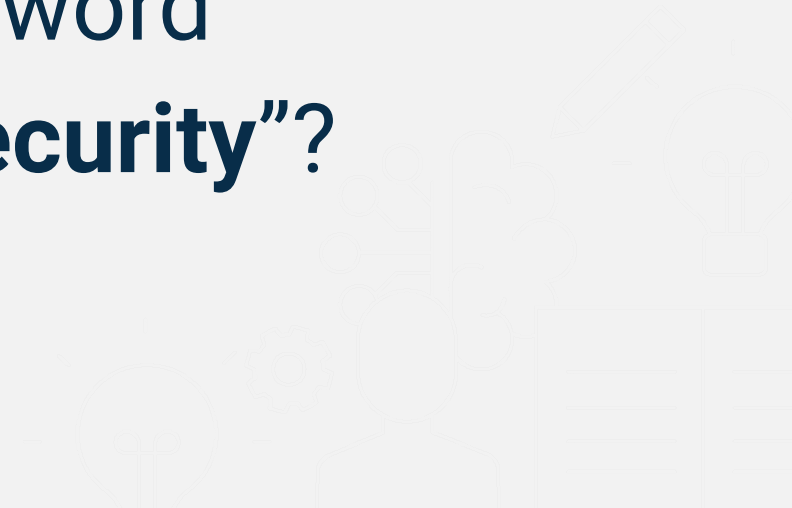
Fileless malware



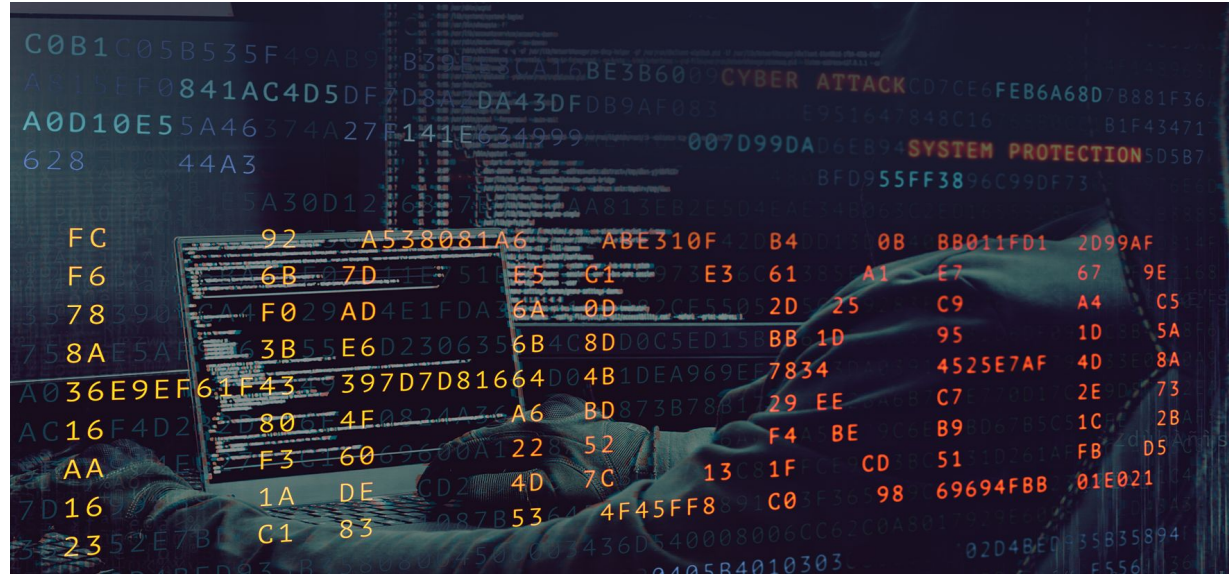
Defining **Cybersecurity**



What do you associate
with the word
“cybersecurity”?



Attackers and complicated code





But cybersecurity isn't
simply about **attackers**
and **complicated code...**



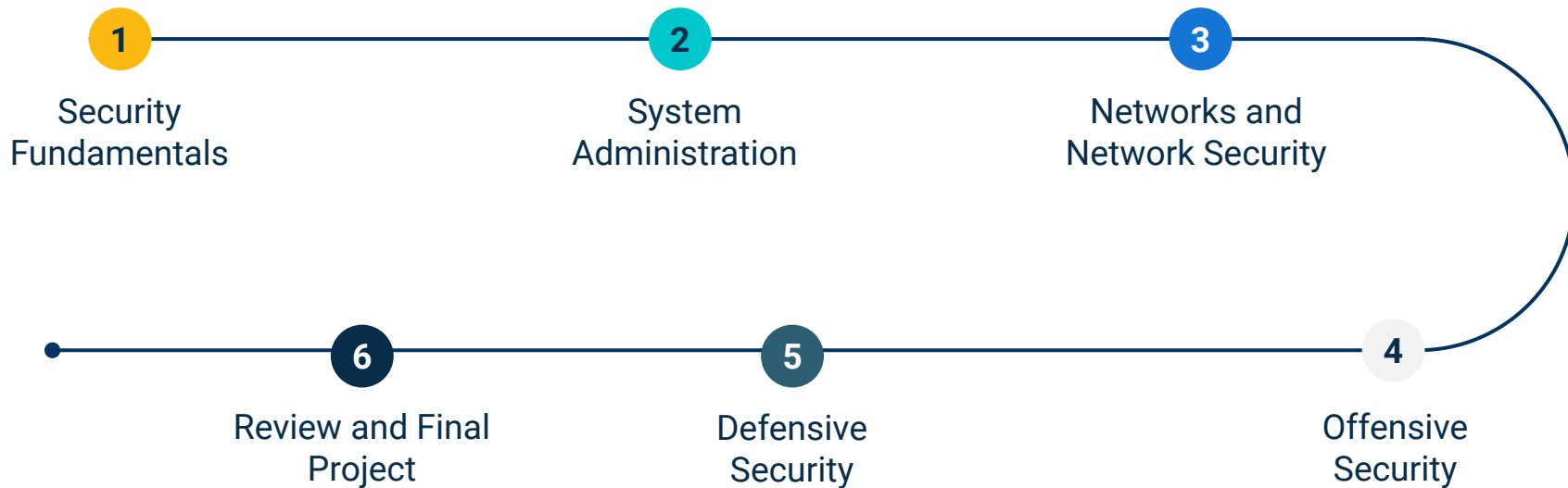


Cybersecurity is really
about **assessing threats**
and **mitigating risks**.



Course **Overview**

Curriculum at a Glance: Units 1–3





Example Assignments

- 1 Using AI tools to generate reports, explain exploits, and analyze logs
- 2 Linux Scavenger Hunt
- 3 Networking Capture the Flag
- 4 Pen testing project week
- 5 Master of the SOC

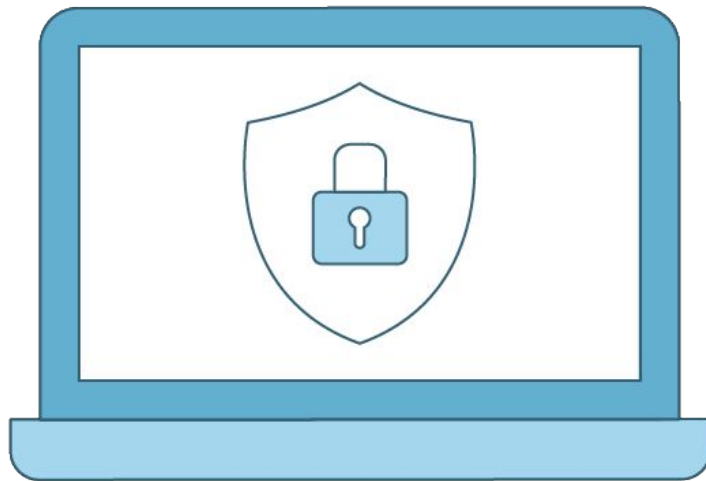


Four Projects:

Project 1

Linux Hardening

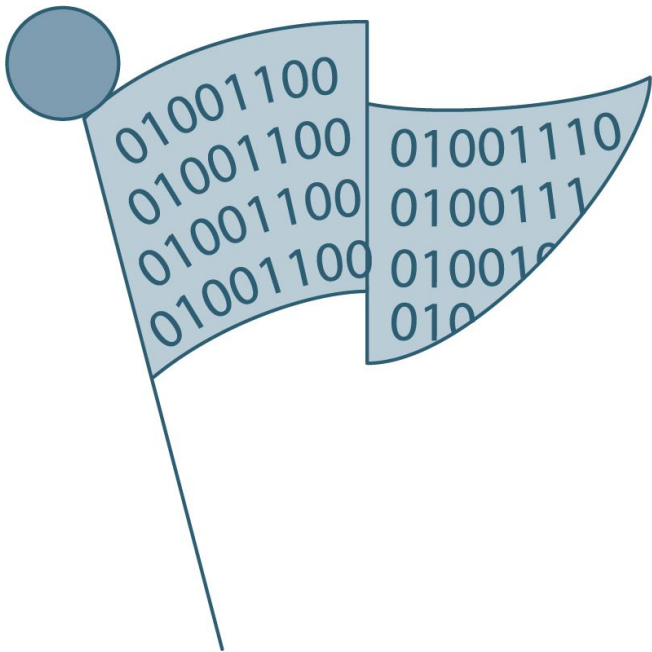
You will play the role of security analysts tasked with auditing and hardening a Linux server for a fictional organization.





Four Projects:

Project 2



Offensive Security CTF

In a three-day CTF, you will use the skills you have learned in the Offensive Security module to attack a fictional organization and determine its vulnerabilities.

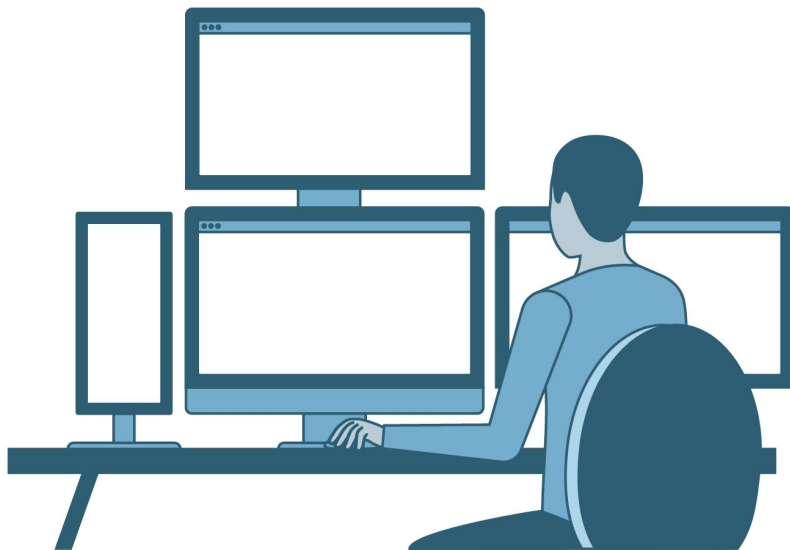


Four Projects:

Project 3

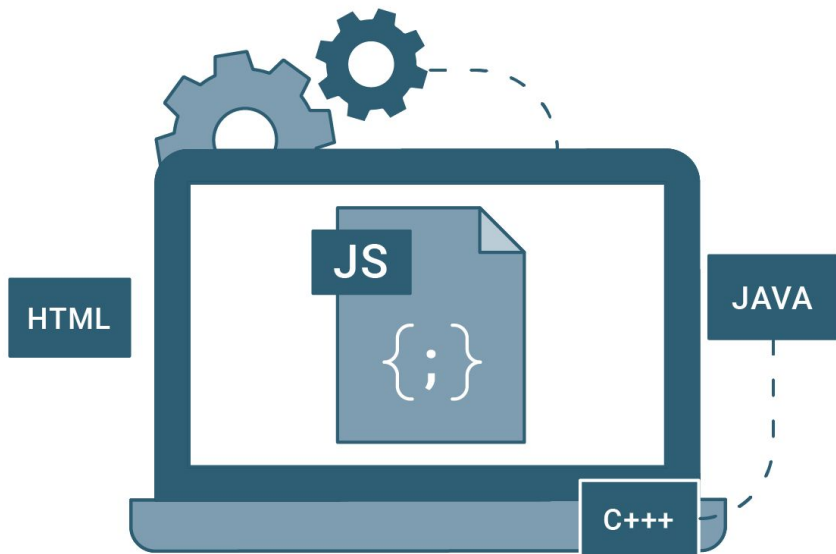
Building a Monitoring Solution

You will design a custom monitoring solution for a fictional organization, monitoring the environment during a series of simulated attacks.





Four Projects:



Project 4

bootCon

Present researched projects demonstrating one of the following:

- Exploiting IoT devices
- Code development
- Cybersecurity tool

Tools We'll Use

We will use **virtual machines** to operate various operating systems and tools throughout the curriculum.



Virtual machines allow us to run different operating systems.



We can download and install virtual machines onto our computers. Or we can access them through web applications.



In cases where we need to use more than one virtual machine, we will access a network of those machines on the cloud.

Student Resources

There are many resources available for learning and review. These include:

01

Student Module Syllabi

These syllabi will contain the weekly overviews of the modules.

They will set expectations and provide additional support.

02

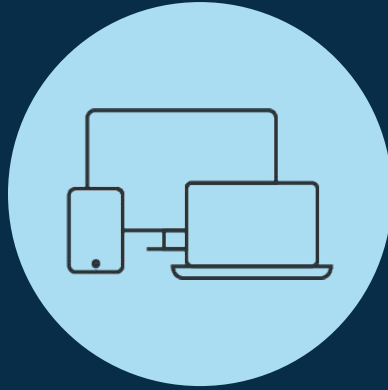
Student Guides

You are encouraged to use these to follow along with the Instructor Demos as well as links to the activities.

03

Slides

Links to the slides used in the presentations are included in the student guides.



Instructor Demonstration

Student Resources & Tools



Online **Learning**



Tips: Online Learning

01

Get to know your classmates and instructors.

You'll get more out of the course if you become part of a shared community.

- Social connection can be difficult to develop through a computer screen, but there are ways to get to know your fellow classmates.
- Participate in class and in your cohort's Slack space.
- You don't have to get too personal, but you can share your specific security interests or career goals.



Tips: Online Learning

02

Engage with class time as you would a live class.

- Make the area around you as distraction-free as possible.
- If you can, go to a quiet room, silence your phone, and ask others in your home to avoid distracting you while you're online.



Tips: Online Learning

03

Budget time for classwork, Challenge assignments, and reviews.

In an online environment, it can be harder to keep track of due dates for assignments.

- Deliberately schedule time throughout the week for your coursework.
- Set aside three hours, three to four times per week, for studying and completing Challenge assignments.
- Your work won't seem as overwhelming this way, and you won't be working on assignments at the last minute!

Best Practices for Zoom

Help make online learning as productive and easy as possible with these guidelines:



01

Always mute.

02

Include your first and last name for your screen name.

03

Keep your video on.

04

Raise your hand.

05

Use headphones with a microphone.

Best Practices for Slack

Slack is an online communication tool that is like a forum, instant messenger, and email all rolled into one. Countless organizations worldwide use this tool, and you'll use it every day for the next six months.



01

Check Slack regularly

We will use Slack to send code snippets during class, share important announcements, and facilitate group exercises.

02

Use your class channel

You should have received the link to your class-specific channel during orientation.

03

Download the desktop app

Though there is a Slack web application, for this course, you should have the program installed on your machine.



Break
15 mins



For the next task, we will
divide into **breakout**
groups and get started
on our first activity!



Activity:

A Wild USB Appears!

In your breakout groups, discuss the following scenario and questions:

When plugged into a computer, the USB drive immediately executes running code.

- How is a USB drive able to do this?
- Why can't our computer stop the drive from running?
- How might we defend against USBs like this?

Suggested Time:
15 Minutes



Time's up!
Let's review



Questions?

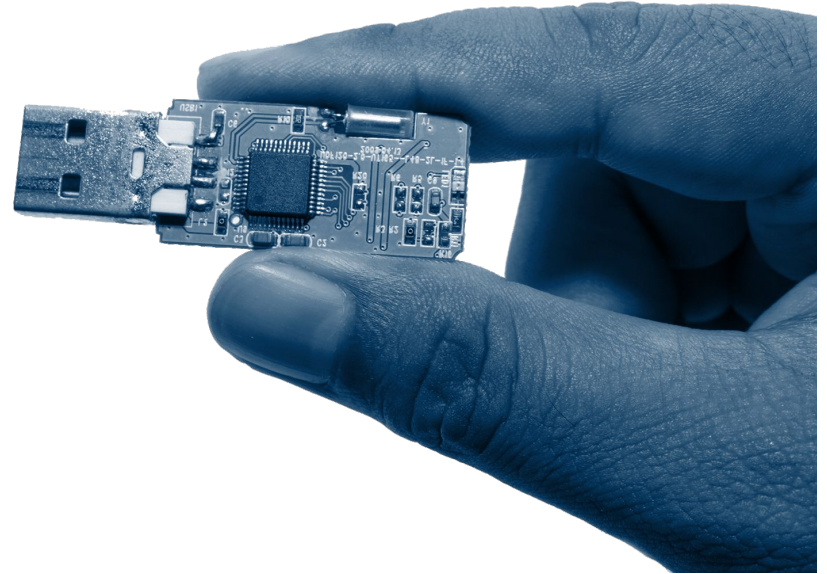


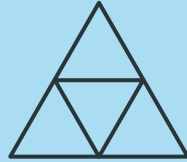
A Harmless USB?

What if the USB was a mini keyboard emulator?

When connected, our computer registers it as a keyboard, allowing it to kick off without restriction.

Like most threats, its appearance is deceptive and seemingly safe.





The **CIA Triad**

The three cornerstones of information security are known as the **CIA triad**.





Confidentiality

The state of being kept secret or private.

This corner of the CIA triad is all about **ensuring that sensitive information does not reach unauthorized people.**

Examples of confidentiality attacks include:

- Uploading private photos and communications onto a forum
- Exposing credit card numbers online



Integrity

The quality of being honest, whole, or undivided.

The integrity of information refers to **protecting information from being modified by unauthorized parties.**

Examples of integrity attacks include:

- Intercepting money transfers and changing the dollar amount in seemingly insignificant ways, so the excess can be sent elsewhere
- Altering university grades to be better or worse



Availability

The quality of being able to be used or obtained.

Availability concerns occur when operating systems, equipment, and data are not functioning correctly, **limiting access** to those who need it.

Examples of availability attacks include:

- Bad actors taking down a web-connected generator and disabling a critical power supply
- Using a denial of service (DoS) attack to bring down a financial service provider's website, making it impossible for clients to make transactions



Activity:

CIA Triad Security Scenario

In this activity, you will analyze a variety of brief security scenarios and identify which element of the CIA triad (confidentiality, integrity, or availability) each situation concerns.

Suggested Time:
15 Minutes





Time's up!
Let's review



Questions?





Next class

We will dive deeper into **assessing risk and mitigating threats** by evaluating specific attacks and vulnerabilities of users, web applications, servers, and databases.



Thank you